

FLASHCARD-ENGINE

Project Documentation & Technical Report

1. Executive Summary

FlashcardEngine is a sophisticated AI-integrated SaaS platform designed to solve the problem of information overload in modern education. Traditional studying methods like passive reading lead to rapid memory decay. This project implements a full-stack solution that automates the creation of study materials from PDFs and utilizes the **SuperMemo-2 (SM-2) algorithm** to ensure long-term retention through spaced repetition.

2. Technical Architecture

Frontend Framework Next.js 14 (App Router) with React, utilizing Client and Server Components for optimal performance.	Styling & UI Tailwind CSS for responsive design and Framer Motion for high-fidelity 3D animations and transitions.
Database & ORM MongoDB Atlas (NoSQL) managed through Prisma ORM to ensure type-safe database interactions.	Authentication Clerk Auth for secure user identity management, session handling, and protected routing.

3. Core Functionality

3.1 AI-Driven Content Generation

The system utilizes Natural Language Processing (NLP) to analyze uploaded PDF documents. It extracts key concepts and generates "Active Recall" question-answer pairs. This automates the most time-consuming part of studying, allowing users to focus entirely on the learning process.

3.2 SM-2 Spaced Repetition Logic

The backbone of the engine is the SM-2 algorithm. For every card reviewed, the system tracks:

- **Repetition (n):** The number of times the card has been successfully recalled.
- **Easiness Factor (EF):** A floating-point number representing how "easy" the card is. It defaults to 2.5 and adjusts based on user feedback.
- **Interval (I):** The number of days until the card is shown again.

```
// EF calculation logic:  
efactor = efactor + (0.1 - (5 - quality) * (0.08 + (5 - quality) * 0.02));  
if (efactor < 1.3) efactor = 1.3;
```

4. Mastery Evaluation System

Unlike simple completion trackers, FlashcardEngine evaluates user knowledge through a mathematical **Mastery Score**. The mastery percentage for a deck is calculated by averaging the Easiness Factors of all included cards and mapping them to a 100% scale.

Mastery Formula:

Mastery % = (Average E-Factor / 2.5) × 100 [Capped at 100%]

5. User Interface Design

- **Dashboard:** A centralized hub displaying real-time analytics, including total library size and average mastery across all subjects.
- **3D Flashcard UI:** A custom-built study interface featuring CSS 3D transforms, perspective rendering, and backface-visibility for a realistic feel.
- **Gamification:** Integration of celebration effects (confetti) upon session completion to provide psychological reinforcement.

6. Conclusion

FlashcardEngine represents a modern approach to education technology. By leveraging cloud databases, AI, and cognitive science, it provides a scalable and efficient way for students to master complex subjects.

Developed By: Bhuvnesh Singh

Academic Focus: B.Tech CSE (Artificial Intelligence & Machine Learning)

Institution: GLA University

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